

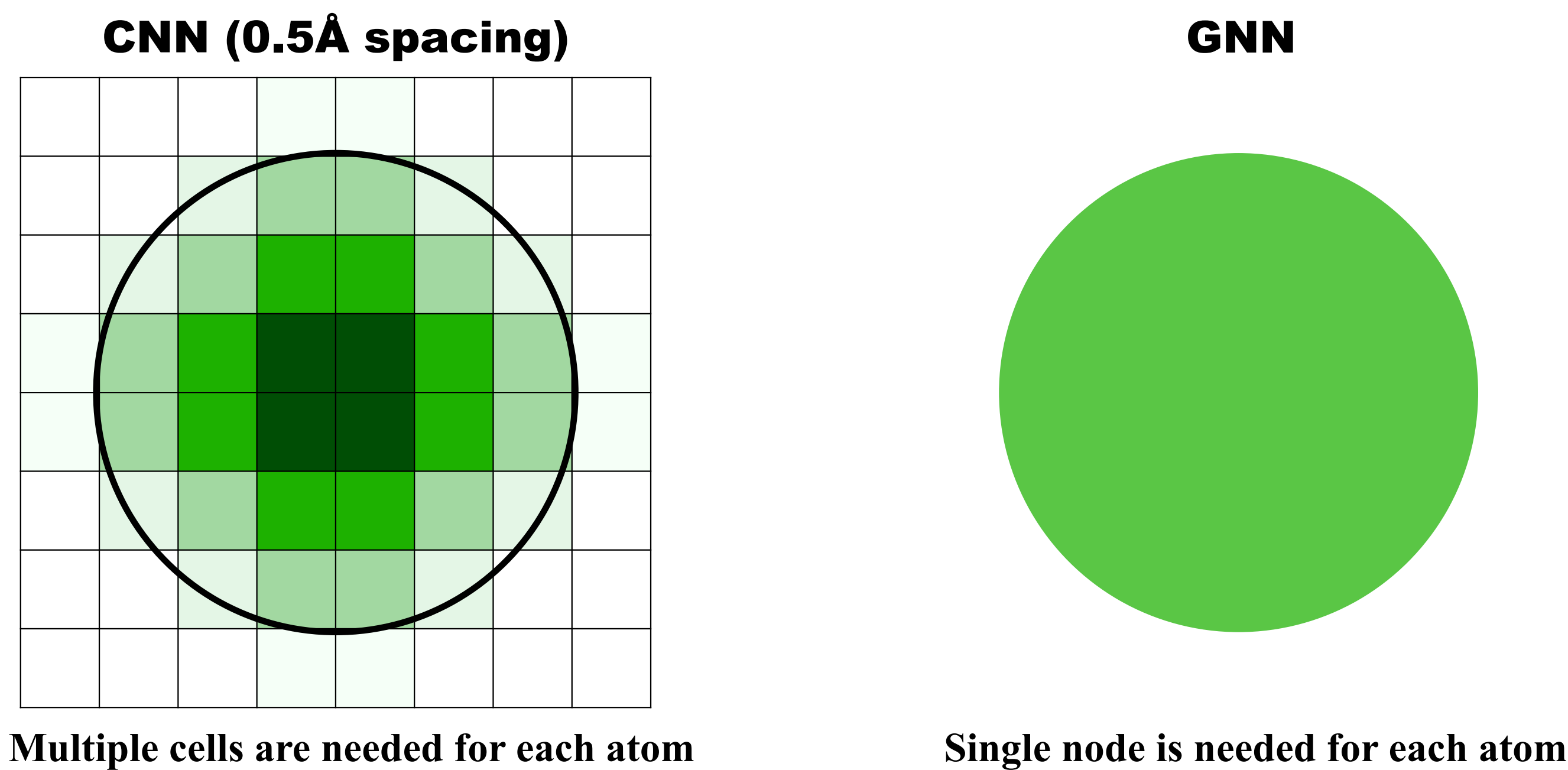
Water position prediction with SE(3)-Graph Neural Network

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ABSTRACT

- Protein-bound **water molecules** affects the structure and function of the protein.
- To consider such water molecules, positions of bound water from given protein structure should be known. If only protein structure is given, **prediction of water molecule's position** could be done.
- For **fast and accurate prediction**, **WatGNN**, water position prediction method using **SE(3)-Graph Neural Network** (SE(3)-GNN) was developed in this research.



- With SE(3)-GNN based method, water position accuracy was better than state-of-art methods, while average computational cost was 1.86 seconds on i9-12900k CPU and RTX-4090 GPU, which is **12.8 times faster** than using our previous prediction method using Convolutional Neural Network.
- The preprint and prediction program is available on Biorxiv and Github, respectively.
- Preprint: www.biorxiv.org/content/10.1101/2024.03.25.586555v1
- Github: <https://github.com/shadow1229/WatGNN> (Github repository)

Preprint

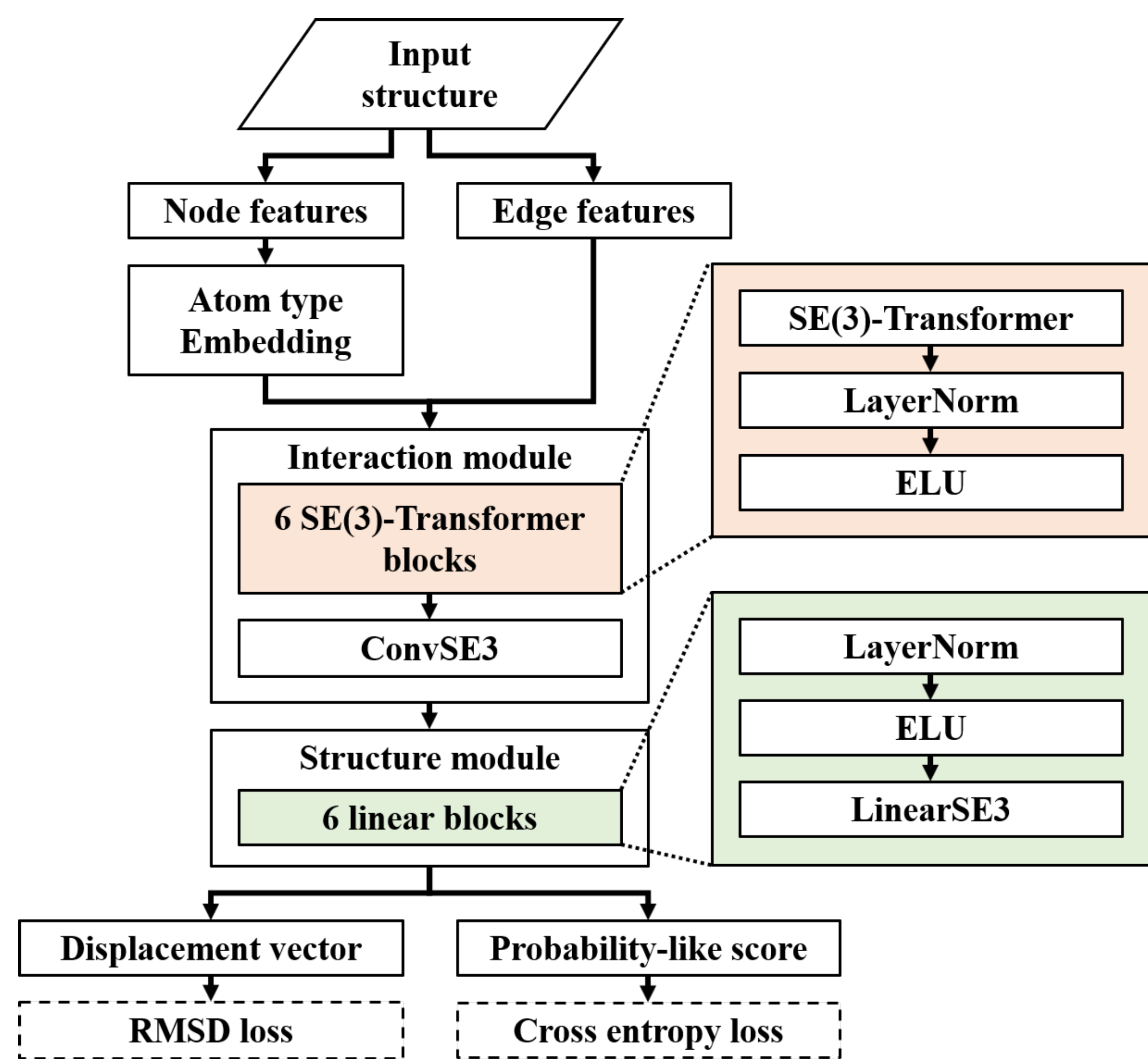


Github



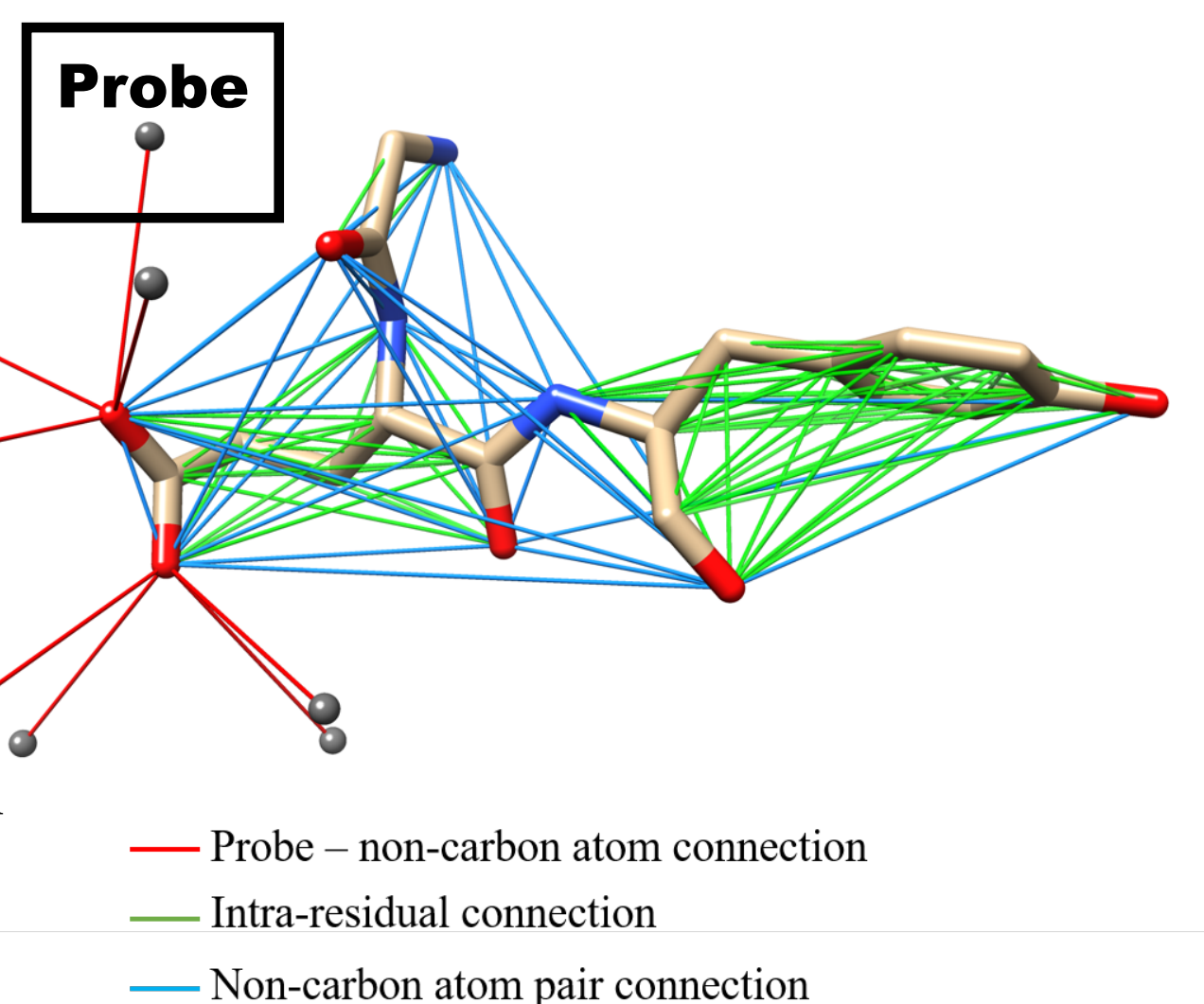
WatGNN Method

Network Structure



Features

- Node features: Atom types only
 - C, N, O, S, P, halogens, alkali metals, non-alkali metals, probes, other atoms
- Edge features
 - Probe-non carbon atom connections
 - Intra-residual connection
 - Non-carbon atom pair connection within 10Å



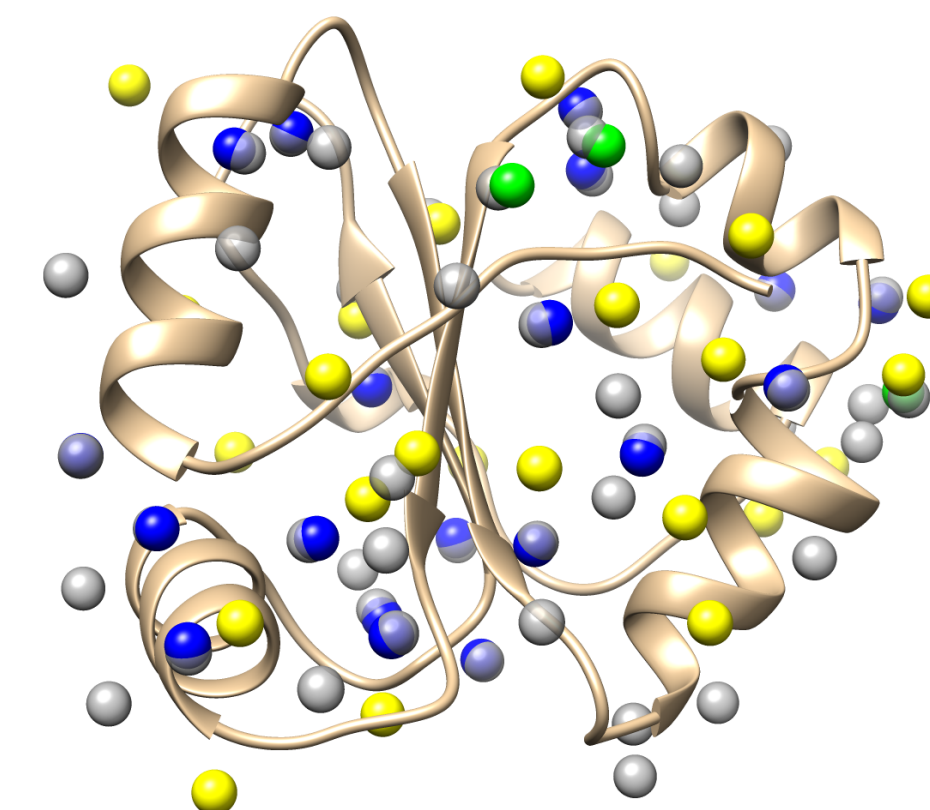
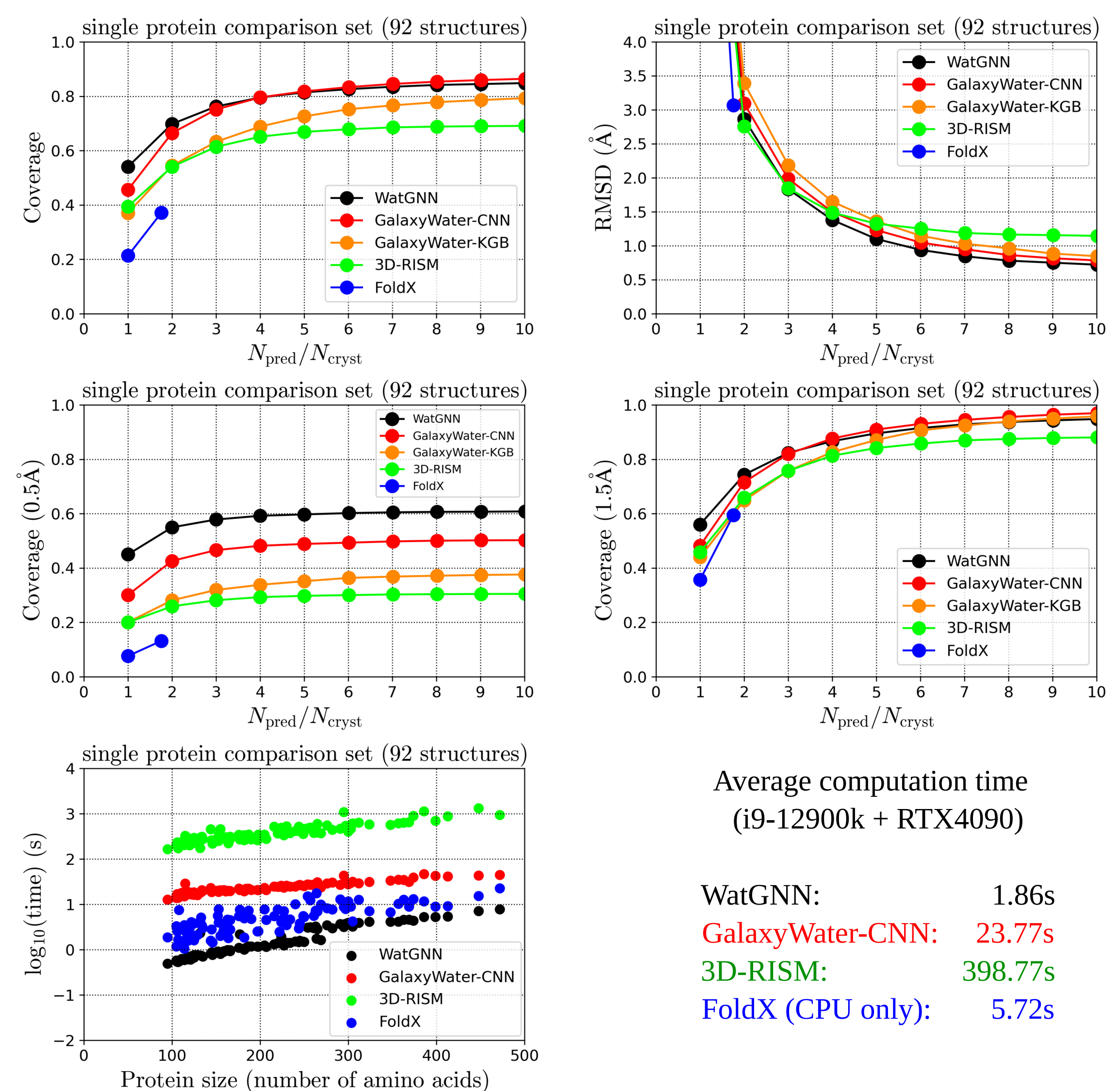
Training set

- High resolution X-ray crystallography protein structure set
 - 2.0Å or better resolution, sequence identity < 30%, < 5000 heavy atoms
- 7971 structures (curation date: Sep 21, 2023)
- training: 3971, validation: 300, test: 3700 structures

RESULT

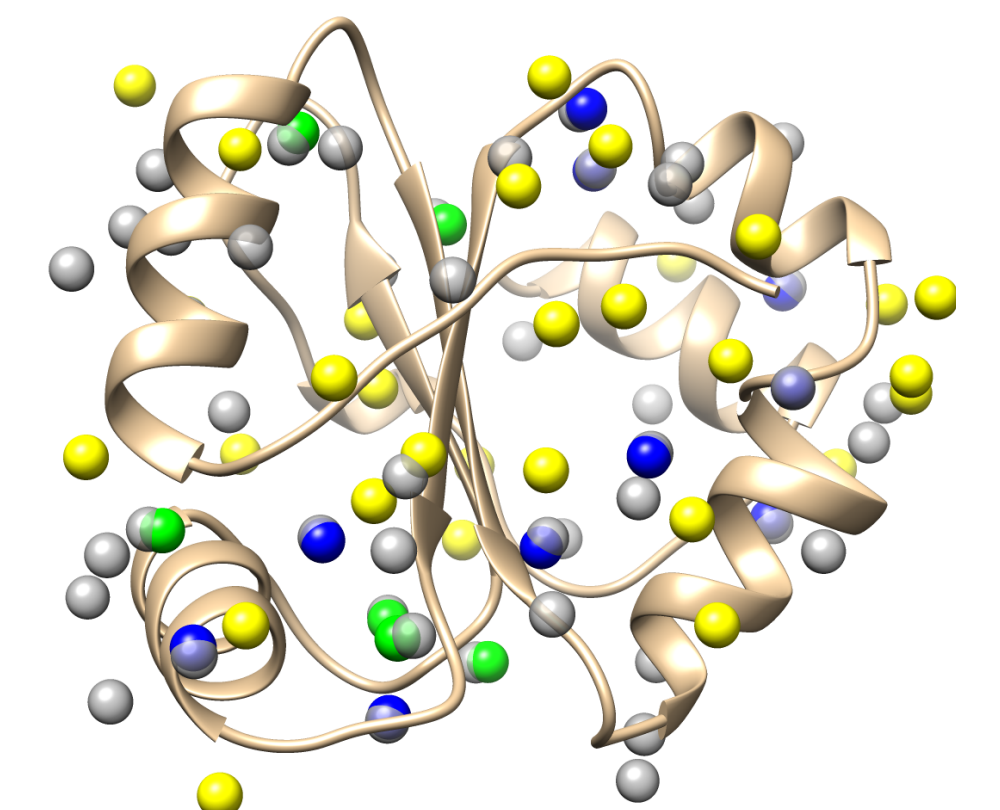
Prediction performance comparison

- Comparison set
 - Single protein structure set
 - 92 X-ray crystal structures, 1Å resolution or better, released before March 14th, 2015.
 - Protein-compound structure set
 - 397 structures curated from the PDDBind set, 2Å resolution or better, sequence identity < 30%, tanimoto similarity < 50%
- Performance for single protein structure set



WatGNN

Predicted water sites



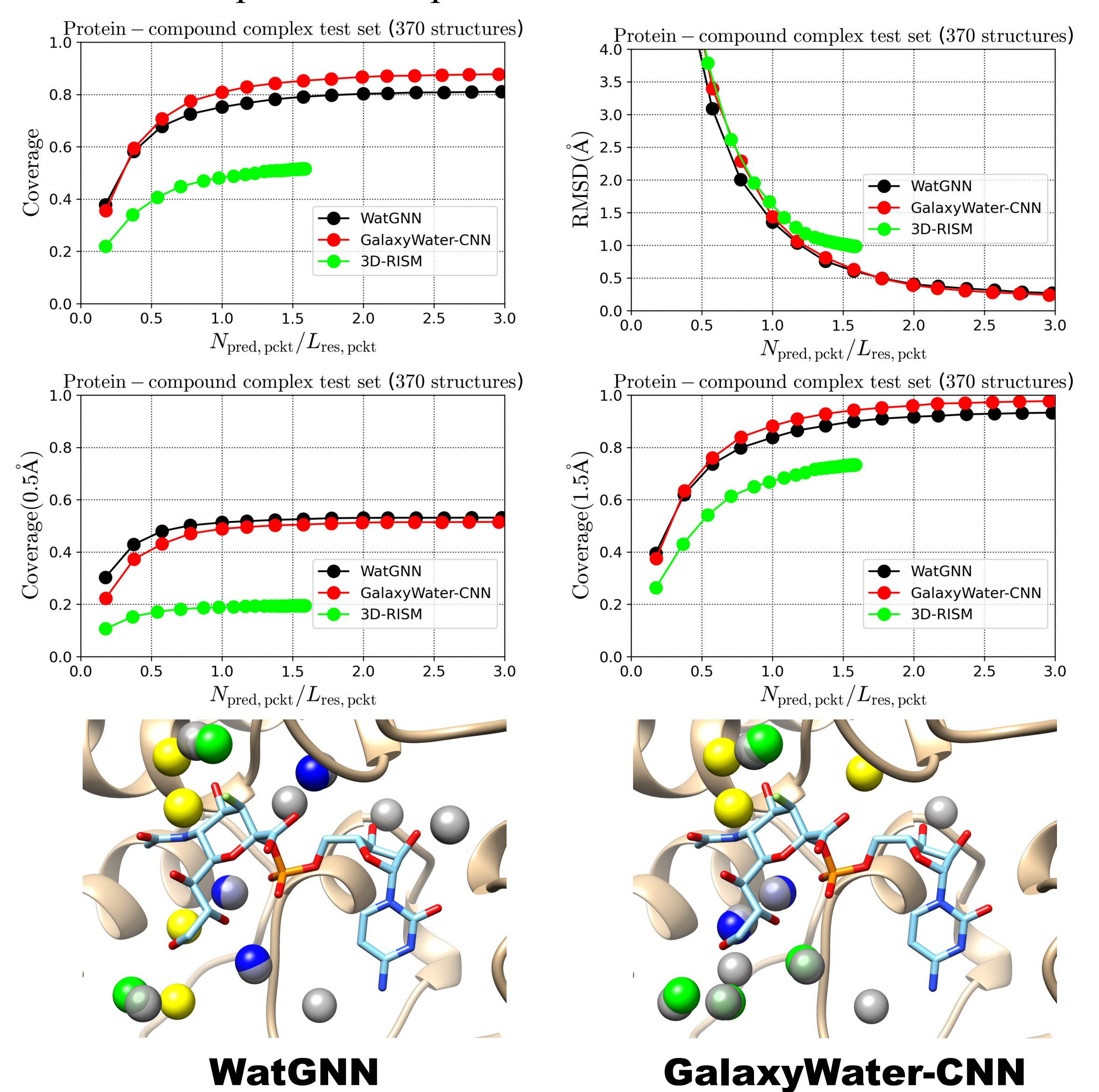
GalaxyWater-CNN

Crystallographic water w/ predicted sites within 0.5Å

Crystallographic water w/ predicted sites within 1.0Å

Crystallographic water w/o predicted sites within 1.0Å

Performance for protein-compound structure set



CONCLUSION

- Considering an atom as a single node for water position prediction via GNN greatly decreases computational cost compared with the previous method using CNN.
- Using GNN enables more accurate water position prediction without limited resolution due to CNN's grid representation.
- The limited amount of protein-ligand complex set for training limits the performance of WatGNN, which implies that the amount of training data impacts performance of WatGNN to what extent.